06 - Text + Image Fusion

Goal: combine the text and image models and check if this beats the best text model.

We compare:

- **Simple fusion:** a weighted average of the SVM and image probabilities, with one fixed weight.
- **Random Forest (stacking):** a second model that learns how to combine the probabilities of CamemBERT, the SVM and the image model (81 numbers per product).

1. Text probabilities

We use the optimised text model from notebook 04 (all features, binary TF-IDF, LinearSVC with C = 1). A LinearSVC gives scores, not probabilities. Calibration learns on the training set (3-fold CV) how to turn these scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1: True
```

2. Match text and image on the same products

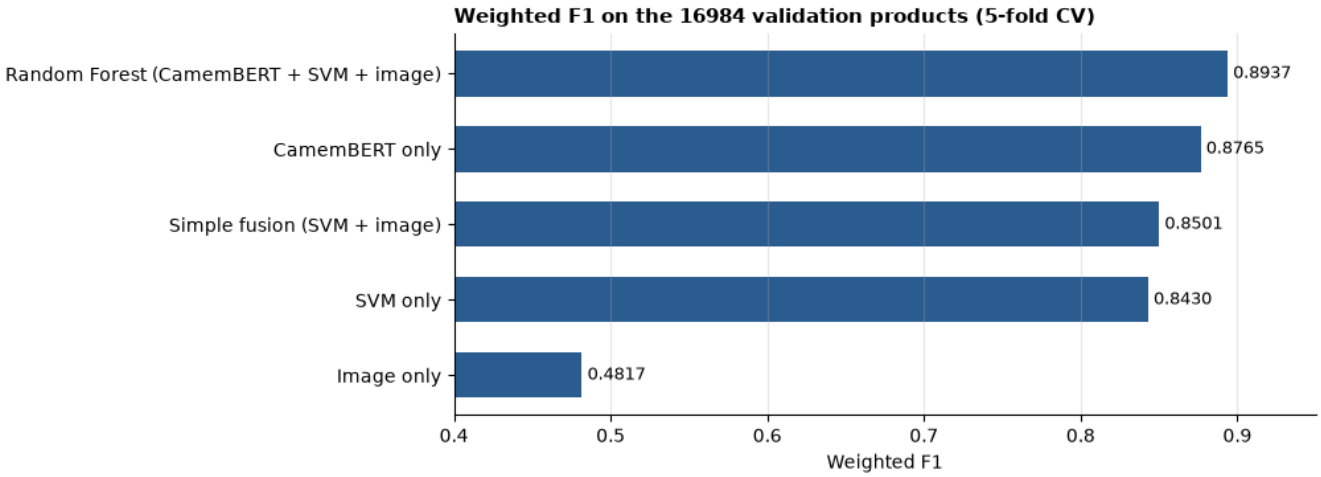
```
Products scored by both: 16984
```

3. Combine the models

Text and image probabilities exist for the 16984 validation products, which none of the models were trained on. We score every method with 5-fold cross-validation on these products: the combiner learns on 4/5 of them and predicts the remaining 1/5. Every product is predicted by a combiner that never saw it.

For simple fusion, the text weight (0 to 1, in steps of 0.1) is chosen on the 4/5 part only, in the same way.

```
Text weight chosen in each fold: [0.7, 0.7, 0.7, 0.7, 0.7]
```



Model	Weighted F1	Macro F1
Random Forest (CamemBERT + SVM + image)	0.8937	0.8801
CamemBERT only	0.8765	0.8594
Simple fusion (SVM + image)	0.8501	0.8401
SVM only	0.8430	0.8326
Image only	0.4817	0.4230

Analysis

- The best model is the Random Forest on CamemBERT, SVM and image: 0.8937 weighted F1 and 0.8801 macro F1.
- That is +1.7 points over CamemBERT alone (0.8765) and +5.1 points over the SVM alone (0.8430).
- Simple fusion with one fixed weight (70% SVM, 30% image) reaches 0.8501: images help, but a fixed weight cannot use them well.
- We also tried Gradient Boosting instead of Random Forest. It scored slightly lower (0.8595 against 0.8630 on SVM + image) and was much slower, so we kept the Random Forest.

4. Per category: best fusion vs CamemBERT

Code	Category	CamemBERT F1	Best fusion F1	Gain
1180	Wargames	0.614	0.691	0.078
2705	New Books	0.818	0.892	0.075
10	Used Books	0.689	0.754	0.065
1302	Outdoor Toys	0.810	0.854	0.045
40	Games	0.795	0.831	0.036
2585	Garden Tools	0.823	0.849	0.026
2220	Pets	0.890	0.915	0.026
60	Consoles	0.876	0.901	0.024
1320	Baby	0.856	0.877	0.021
2582	Garden Furniture	0.794	0.815	0.020
1940	Food	0.946	0.966	0.020
2462	Used Games	0.821	0.838	0.017
1160	Cards	0.955	0.971	0.016
2522	Stationery	0.945	0.959	0.015
2583	Pool	0.970	0.982	0.012
1140	Figurines	0.834	0.846	0.012
1280	Toys	0.791	0.803	0.012
1920	Textiles	0.925	0.934	0.009
2060	Decor	0.861	0.870	0.009
50	Game Accessories	0.880	0.889	0.009
2403	Book Lots	0.868	0.873	0.005
1560	Furniture	0.882	0.887	0.005
1281	Board Games	0.684	0.689	0.005
2280	Magazines	0.919	0.923	0.003
1300	RC Models	0.987	0.987	0.000
2905	PC Games	0.991	0.991	0.000
1301	Baby Socks	0.978	0.975	-0.003

Analysis

- The best fusion is better than CamemBERT on 26 of the 27 categories. Only Baby Socks drops slightly (-0.3 points).
- The largest gains are on Wargames (+7.8 points), New Books (+7.5) and Used Books (+6.5).
- Wargames was CamemBERT's weakest point. The SVM catches its rare exact words (*warhammer*, *heroclix*), so combining both fixes it.
- Books gain from both the SVM and the image.

Conclusion

- Combining models helps, and a learned combination (Random Forest) works better than a fixed weight.
- Best model: Random Forest on the CamemBERT, SVM and image probabilities, with 0.8937 weighted F1 on the validation products (5-fold CV).
- Each part adds something: CamemBERT reads context, the SVM catches rare specific words, and the image helps on books.